Reward functions, on one page

Teach a Small AI to Think · unicodesnwrites.tech · one-page cheat sheet

A reward function is a small program that looks at a model's answer and gives it a score. In reinforcement learning, that score is the only feedback the model gets. It never sees the "right answer", only how well it did.

The two rewards behind R1-style training

Reward	Question it asks	Typical score	How to check
Format	Is the answer shaped like `<think>…</think><answer>…</answer>`?	1.0 or 0.0 (or partial credit)	A regular expression, plus "each tag exactly once"
Correctness	Is the final answer right?	1.0 or 0.0	Pull out the number, compare to the gold answer (GSM8K: the number after `####`)

Make correctness worth more than format (for example 1.0 vs 0.5), so being right always beats being tidy.

Verifiable vs judged rewards

	Verifiable (rule-based)	Judged (reward model / AI judge)
Examples	Maths answers, unit tests, exact-match puzzles	Helpfulness, style, essays
Cost	Almost free	Needs another model
Hard to fool?	Mostly, if written carefully	Easier to game at scale
Used by R1 for reasoning?	Yes	No (DeepSeek-AI, 2025, cited reward hacking risk)

The shape TRL expects

```
def my_reward(completions, answer, **kwargs):   # dataset columns arrive by name
    return [1.0 if ... else 0.0 for c in completions]   # one float per completion
```

- `completions` is a list of strings, or for chat datasets a list of messages: use `c[0]["content"]`.
- Always accept `**kwargs`: the trainer passes extra arguments.
- Several reward functions are **added together** (or weighted with `reward_weights`).
- Return `None` for a sample to say "this reward does not apply here".

Checklist before you train

Check	Why
Test on hand-written good, bad and weird answers	Find loopholes before the model does
Missing tags return 0, never crash	Early answers are messy
Numbers like `$1,200.` and `1200` both parse	Grade the number, not the decoration
Each tag must appear exactly once	Stops "repeat the answer block" tricks
Empty `<think></think>` is not rewarded (if you care)	Stops "skip the thinking" tricks
Format worth less than correctness	Stops "tidy nonsense"
Read 10 real samples every few hundred steps	Curves can look fine while answers are rubbish

Reward hacking: signs and fixes

What you see	Likely cause	Fix
Reward climbs, real accuracy flat	The reward measures something easier than the goal	Tighten the check; add a held-out accuracy eval
Answers get very short	Format rewarded, thinking not needed	Lower the format weight; reward correctness more
Same phrase repeated	A rule rewards a keyword or tag count	Count tags exactly once; remove keyword rewards
Many numbers in the answer	Grader accepts "any number matches"	Grade only the last (or only) number in `<answer>`
Reward stuck at 0	Reward too strict for a beginner model	Add partial credit (soft format) to get learning started

> **Goodhart's law:** when a measure becomes a target, it ceases to be a good measure. Your reward is a measure. Training makes it the target.

Further reading: DeepSeek-R1 (DeepSeek-AI, 2025); "Concrete Problems in AI Safety" (Amodei et al., 2016); "Scaling Laws for Reward Model Overoptimization" (Gao et al., 2022); TRL GRPOTrainer docs.